



Technical University of Cluj - Napoca
Computer Science Department

Procesarea Imaginilor

Curs 7:

Convolutie. Transformata Fourier



Notiuni preliminare

Imagine “continua” := functie continua de 2 variabile independente

Exemple: $u(x, y)$; $v(x, y)$; $f(x, y)$

Functie 2D separabila:

$$f(x, y) = f_1(x)f_2(y)$$

Imagine discreta (digitala) := secventa 2D (bidimensionala) de numere reale/intregi

Exemple: $u_{m,n}$; $v(m, n)$

Notatii:

k, l, m, n – indici intregi folositi in matrice, vectori:

$$\mathbf{j} = \sqrt{-1}$$



Convolutie

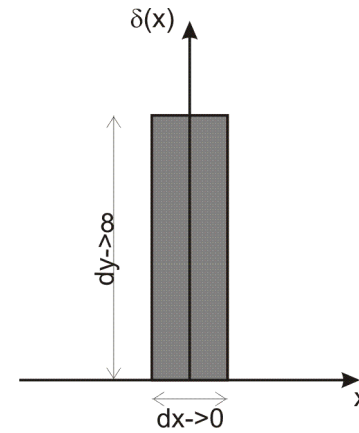
Funcția Dirac (impuls unitar $\Rightarrow$ Aria = 1)

Funcția Dirac 2D (continua):

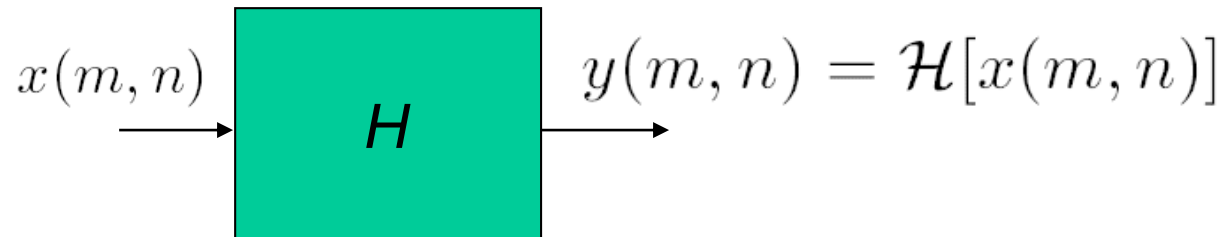
$$\delta(x,y) = \delta(x)\delta(y)$$

Funcția discretă – Kronecker:

$$\delta(m,n) = \delta(m)\delta(n)$$



Sistem



Sistem liniar (principiul superpoziției liniare):

$$\begin{aligned}\mathcal{H}[a_1x_1(m,n) + a_2x_2(m,n)] &= a_1\mathcal{H}[x_1(m,n)] + a_2\mathcal{H}[x_2(m,n)] \\ &= a_1y_1(m,n) + a_2y_2(m,n)\end{aligned}$$

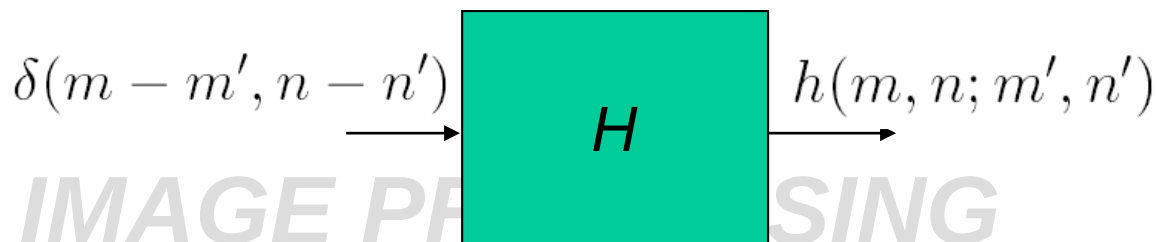


Iesirea unui sistem liniar

$$\begin{aligned} y(m, n) &= \mathcal{H}[x(m, n)] \\ &= \mathcal{H} \left[\sum_{m'} \sum_{n'} x(m', n') \delta(m - m', n - n') \right] \\ &= \sum_{m'} \sum_{n'} x(m', n') \mathcal{H} [\delta(m - m', n - n')] \end{aligned}$$

$$y(m, n) = \sum_{m'} \sum_{n'} x(m', n') h(m, n; m', n')$$

Unde: $h(m, n; m', n')$ Este **raspunsul la impuls al sistemului** – iesirea sistemului H in punctul/locatia (m, n) atunci cand la intrare se aplica functia impuls unitar (Kronecker) la locatia (m', n') .

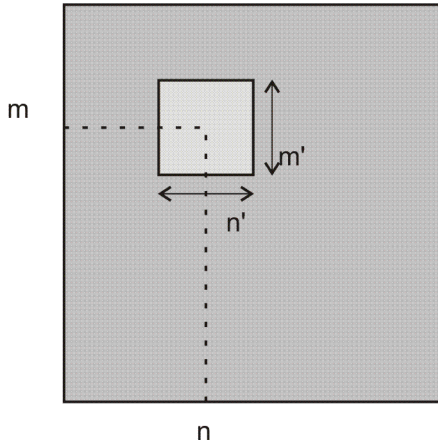


Dc. $y(m, n), x(m, n) > 0$ (ex: imagini), atunci h s.n. **Points Spread Function (PSF)**



Convolutie

Regiune suport a raspunsului la impuls := cea mai mica regiune din planul (m,n) cu proprietatea ca in afara acestei regiuni raspunsul la impuls (h) este nul.



Sisteme **FIR** – sisteme a caror raspuns la impuls au o regiune suport **finita**

Sisteme **IIR** – sisteme a caror raspuns la impuls au o regiune suport **infinita**

Sistem invariant la translatie := sistem pt. care o translatie a intrarii va determina o translatie corespunzatoare a iesirii

⇒ forma raspunsului la impuls nu se schimba odata cu deplasarea impulsului in planul (m,n)

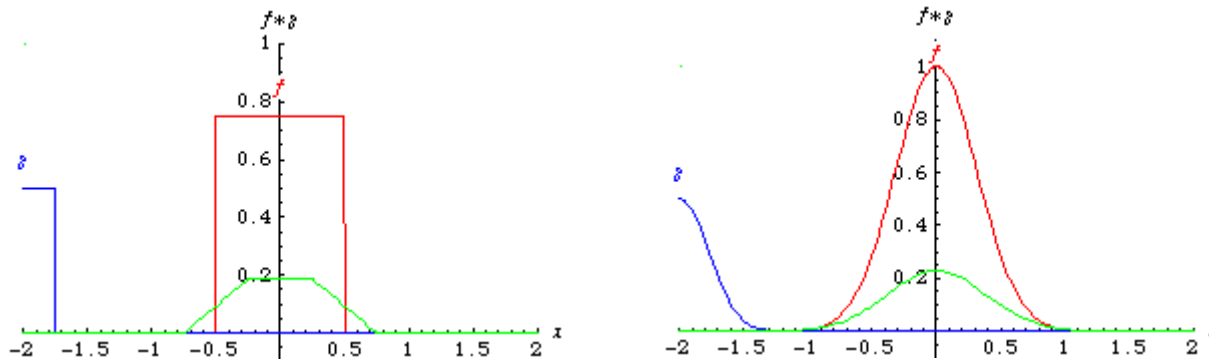
Convolutia intarii cu raspunsul la impuls:

$$y(m, n) = \sum_{m'=-\infty}^{\infty} \sum_{n'=-\infty}^{\infty} h(m - m', n - n') x(m', n')$$



Convolutie

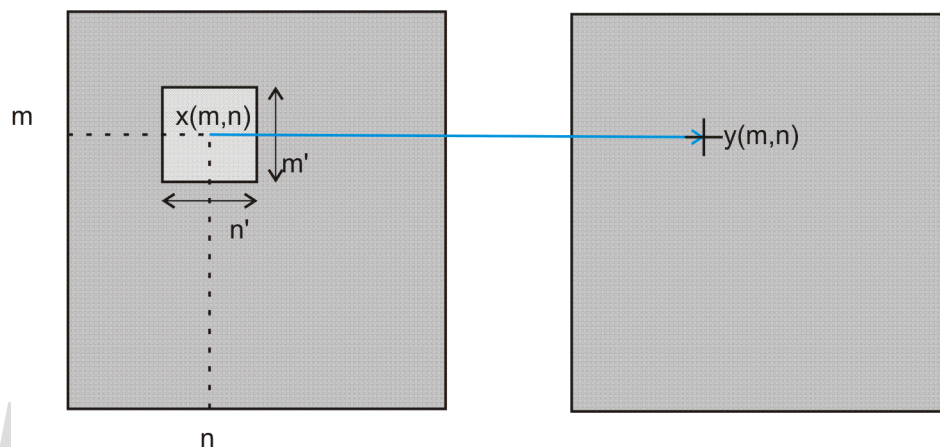
Convolutie 1D



Convolutie 2D

$$g(x, y) = h(x, y) * f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x - x', y - y') f(x', y') dx' dy'$$

$$y(m, n) = h(m, n) * x(m, n) = \sum_{m'=-\infty}^{\infty} \sum_{n'=-\infty}^{\infty} h(m - m', n - n') x(m', n')$$



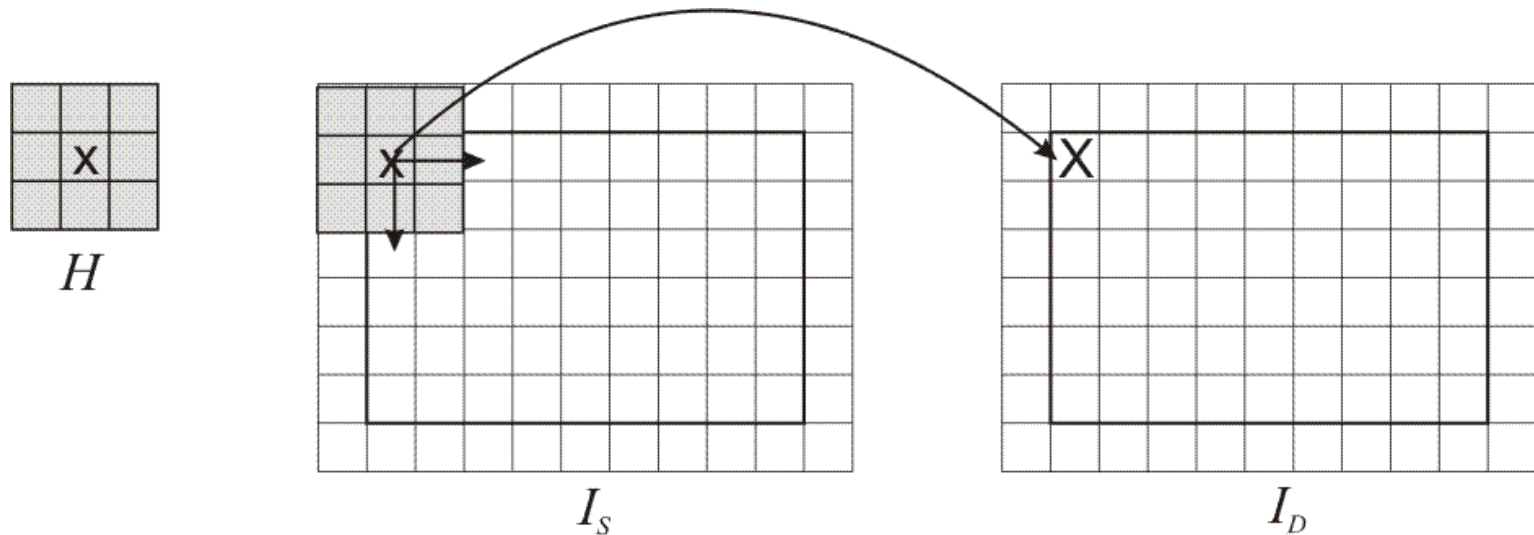


Operația de convoluție în domeniul spațial

Operația de convoluție implică folosirea unei măști/nucleu de convoluție H (de obicei de formă simetrică de dimensiune $w \times w$, cu $w=2k+1$) care se aplică peste imaginea sursă:

$$I_D = H * I_S$$

$$I_D(x, y) = \sum_{i=-k}^k \sum_{j=-k}^k H(i, j) \cdot I_S(x+i, y+j), \quad x = 0 \dots \text{Height} - 1, \quad y = 0 \dots \text{Width} - 1$$





Convolutie

Exemplu: filtru pt. detectia muchilor verticale (calculeaza derivata imaginii pe directia x)

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

ImgSrc (i,j)



$$\text{ImgDst}(i,j) = G_x * \text{ImgSrc}(i,j)$$





Transformata Fourier [1]

Notiuni preliminare

Functie periodica $\Rightarrow$ **serii Fourier** (suma sin si cos de frecvente diferite)

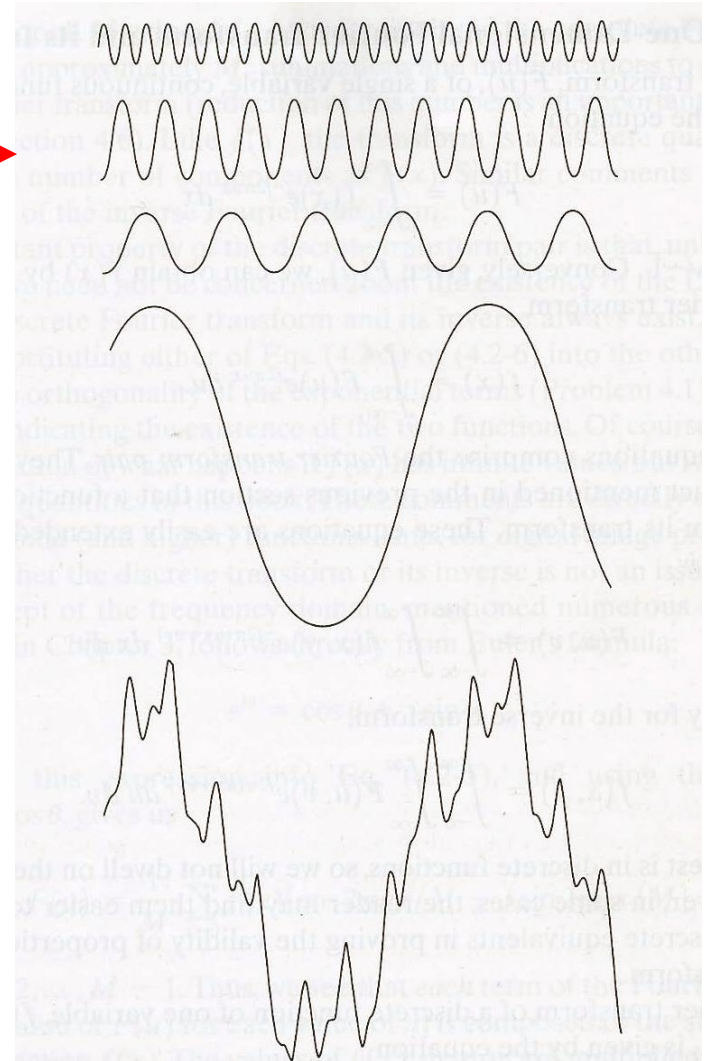
Functie integrala (arie) finita $\Rightarrow$ **transformata Fourier** (integrala / suma de sin si cos)

Transformata Fourier (1D)

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-j2\pi ux} dx$$
$$e^{jx} = \cos(x) + j \sin(x)$$

Transformata Fourier inversa (1D)

$$f(x) = \int_{-\infty}^{\infty} F(u) e^{j2\pi ux} du$$



$\Rightarrow$ Se poate reface functia initiala plecand de la functia transformata !!!



Transformata Fourier [1]

Transformata Fourier (2D):

$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux+vy)} dx dy$$

$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux+vy)} du dv$$

Transformata Fourier discreta - DFT (1D):

$$F(u) = \frac{1}{M} \sum_{x=0}^{M-1} f(x) e^{-j2\pi ux/M} \quad u = 0, 1, 2, \dots, M-1 \text{ (domeniu de frecvente)}$$

$$F(0) = \dots, F(1) = \dots, \dots, F(M-1) = \dots \text{ (componente de frecventa)}$$

$$f(x) = \sum_{u=0}^{M-1} F(u) e^{j2\pi ux/M} \quad x = 0, 1, 2, \dots, M-1 \text{ (domeniu spatial)}$$

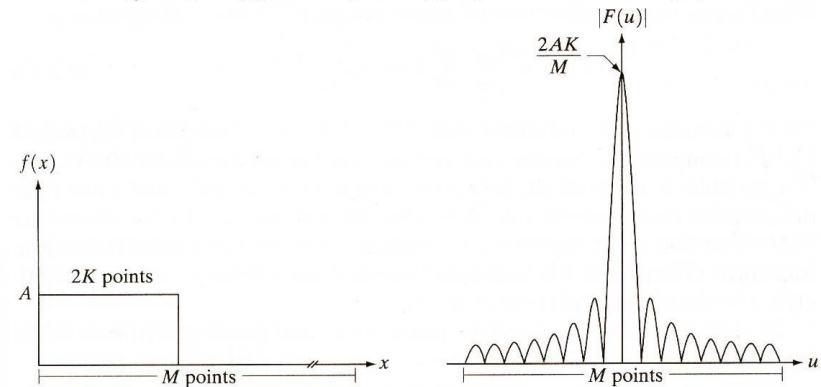
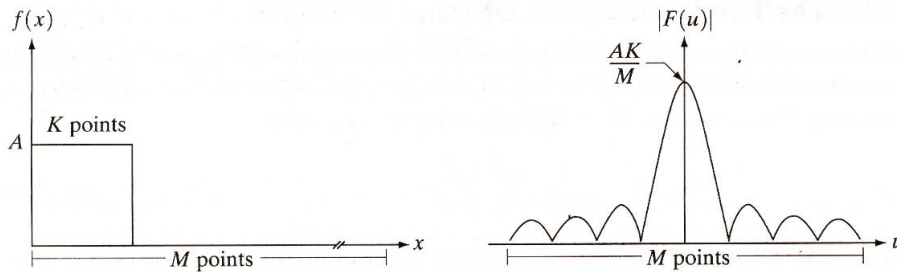


Transformata Fourier [1]

Caracteristici ale DFT

$F(u)$ exprimata in coordonate polare: $F(u) = |F(u)|e^{-j\phi(u)}$

Magnitudine (spectru) $\Rightarrow$ **image enhancement** $|F(u)| = [R^2(u) + I^2(u)]^{1/2}$



Faza (spectru de faza / unghi de faza)

$$\phi(u) = \tan^{-1} \left[\frac{I(u)}{R(u)} \right]$$

Spectru de putere (densitate spectrala)

$$P(u) = |F(u)|^2 = R^2(u) + I^2(u)$$



Transformata Fourier [1]

Transformata Fourier discreta – DFT (2D)

$$F(u, v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

$u = 0, 1, 2, \dots, M-1, v = 0, 1, 2, \dots, N-1$ (coordonate/varibile de **frecventa**)

$$f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$$

$x = 0, 1, 2, \dots, M-1, y = 0, 1, 2, \dots, N-1$ (coordonate/varibile **spatiale**)

Spectru: $|F(u, v)| = [R^2(u, v) + I^2(u, v)]^{1/2}$

Faza: $\phi(u, v) = \tan^{-1} \left[\frac{I(u, v)}{R(u, v)} \right]$

Spectru de putere: $P(u, v) = |F(u, v)|^2$
 $= R^2(u, v) + I^2(u, v)$

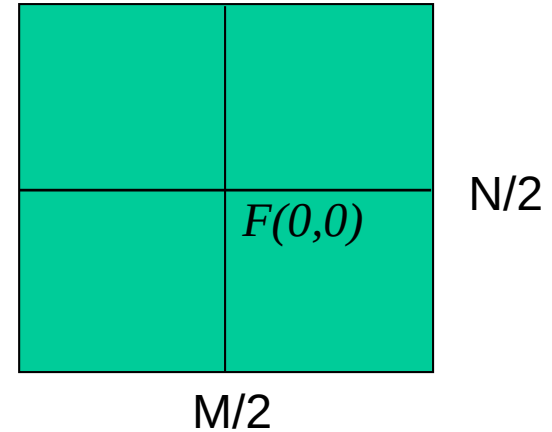


Transformata Fourier [1]

“Shift”-area transformatei Fourier:

$$\mathfrak{S}[f(x, y)(-1)^{x+y}] = F(u - M/2, v - N/2)$$

$\Rightarrow$ Spectru Fourier $F(u, v)$ centrat



Componenta continua a spectrului:

$$F(0, 0) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \Rightarrow \text{intensitatea medie a imaginii}$$

Pt. Imagini ($f \in \mathbb{R}$): $|F(u, v)| = |F(-u, -v)|$

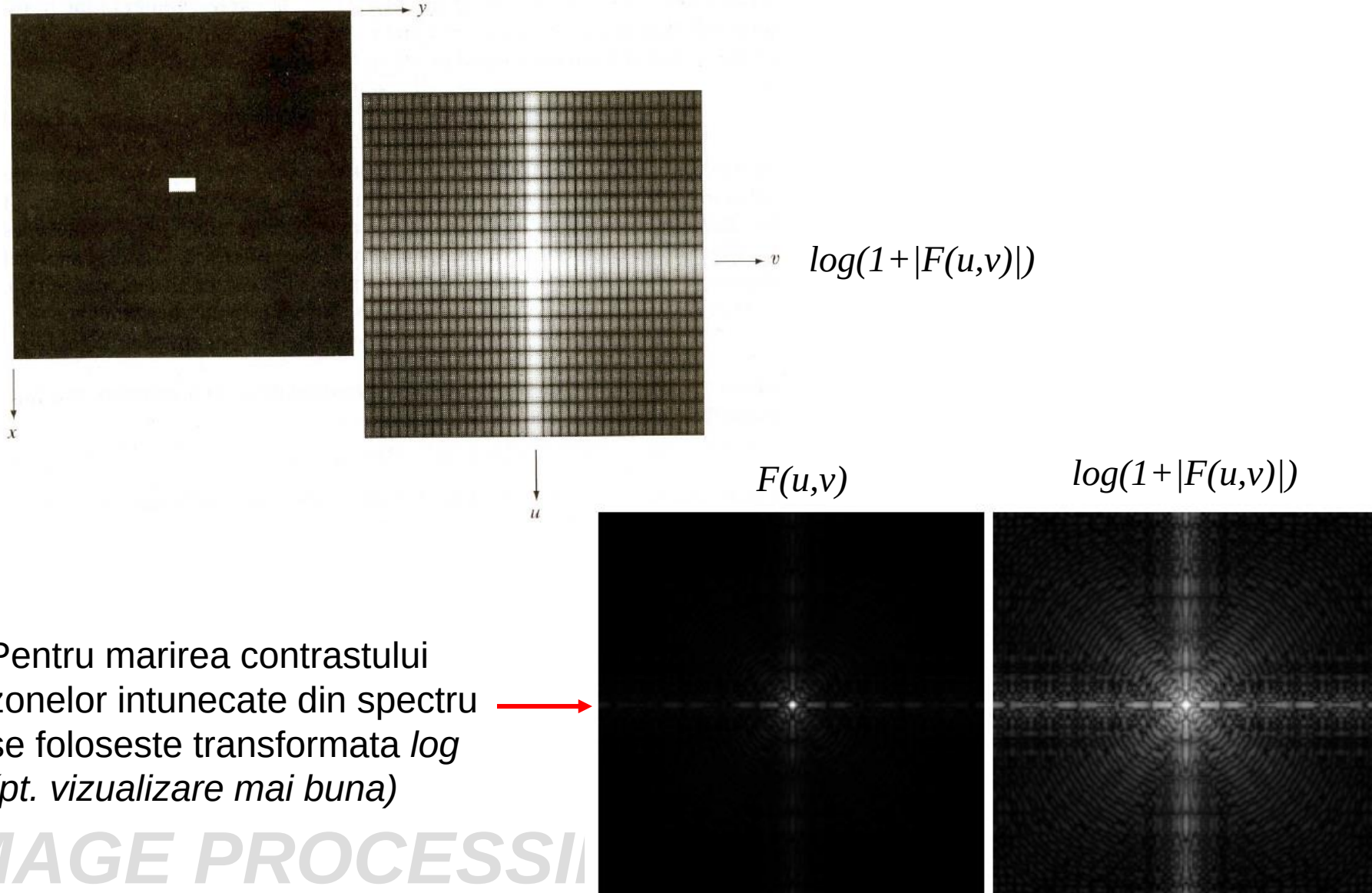
Relatiile dintre esantioanele spatiale si frecventiale ale imaginii:

$$\Delta u = \frac{1}{M\Delta x} \quad \Delta v = \frac{1}{N\Delta y}$$



Transformata Fourier [1]

Exemple de reprezentare a DFT



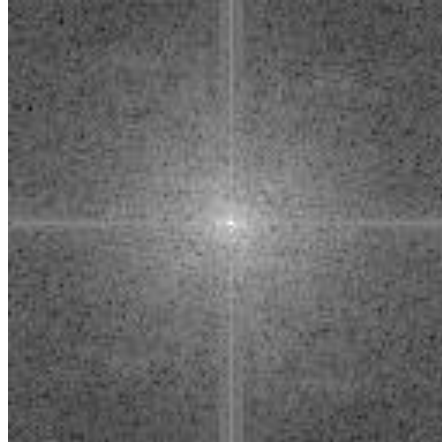


Transformata Fourier

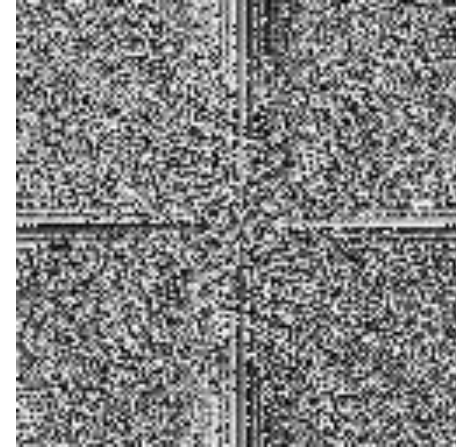
Transformata directa



Original

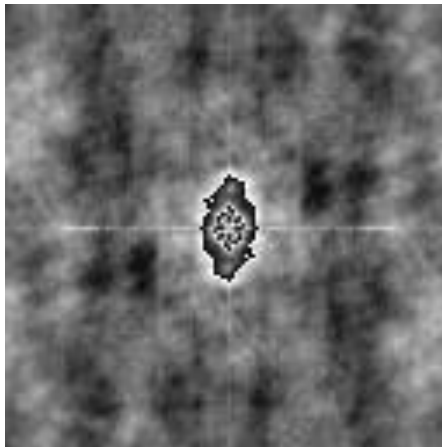


$\log(|F(u,v)|)$



$\phi(u,v)$

Transformata inversa



$\phi(u,v)=0$ (se ignora faza)



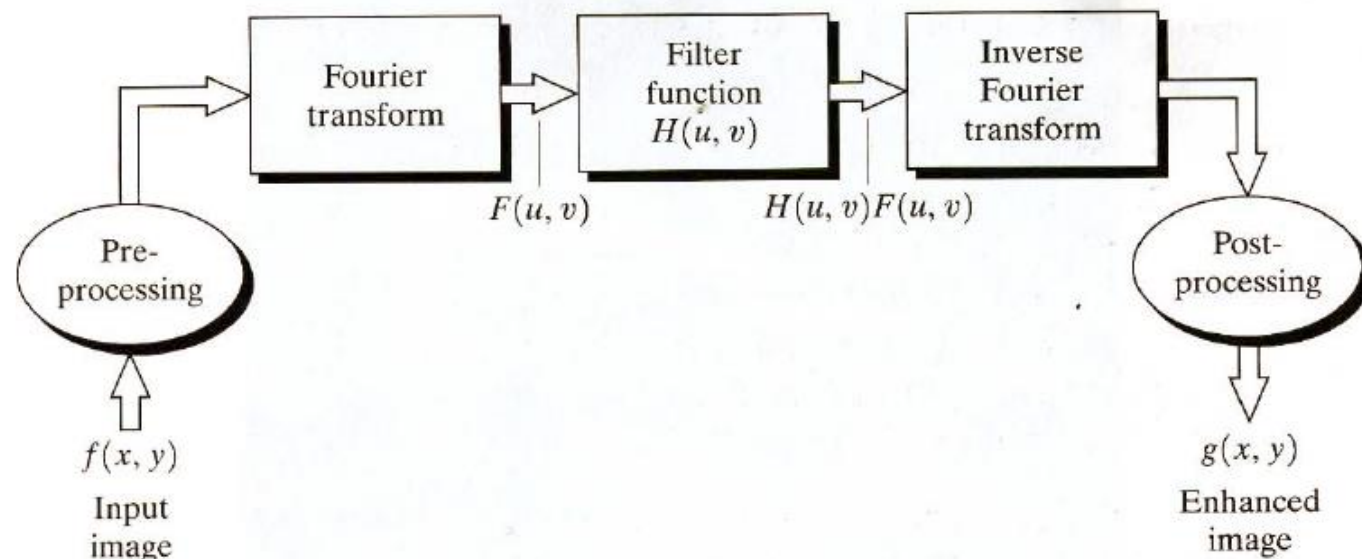
$|A(u,v)|=0$ (se ignora amplitudinea)



Transformata Fourier [1]

Filtrarea imaginilor in domeniul frecventelor $f(x, y) * h(x, y) \Leftrightarrow F(u, v)H(u, v)$

$$f(x, y)h(x, y) \Leftrightarrow F(u, v) * H(u, v)$$

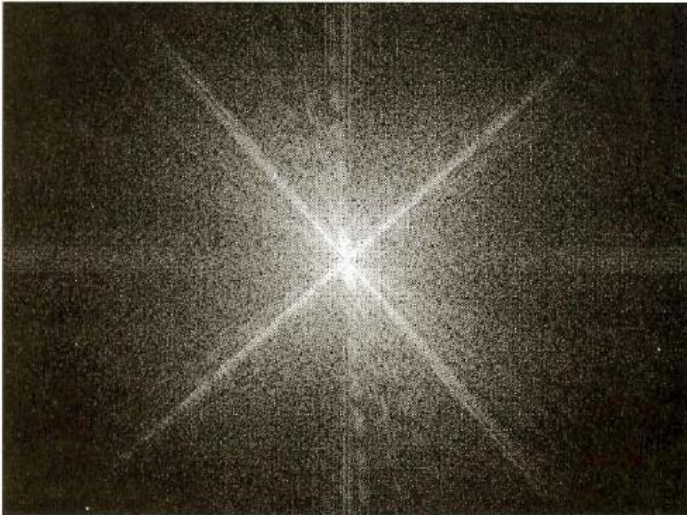
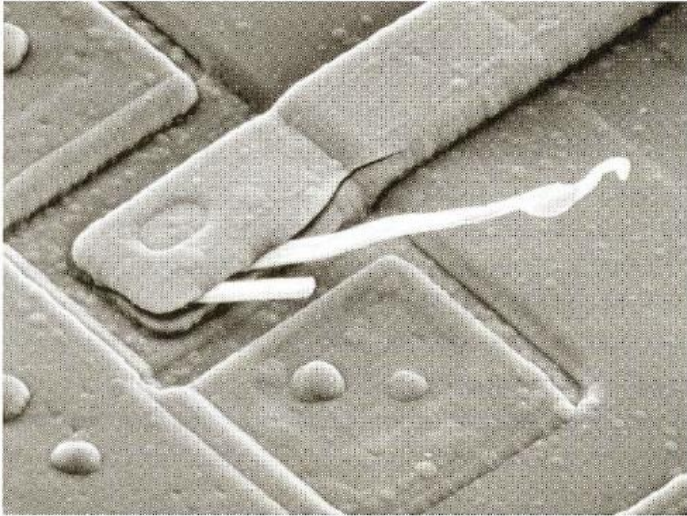


1. Multiply the input image by $(-1)^{x+y}$ to center the transform, as indicated in Eq. (4.2-21).
2. Compute $F(u, v)$, the DFT of the image from (1).
3. Multiply $F(u, v)$ by a *filter* function $H(u, v)$. $G(u, v) = H(u, v)F(u, v)$
4. Compute the inverse DFT of the result in (3). $\text{Filtered Image} = \mathfrak{F}^{-1}[G(u, v)]$
5. Obtain the real part of the result in (4).
6. Multiply the result in (5) by $(-1)^{x+y}$.



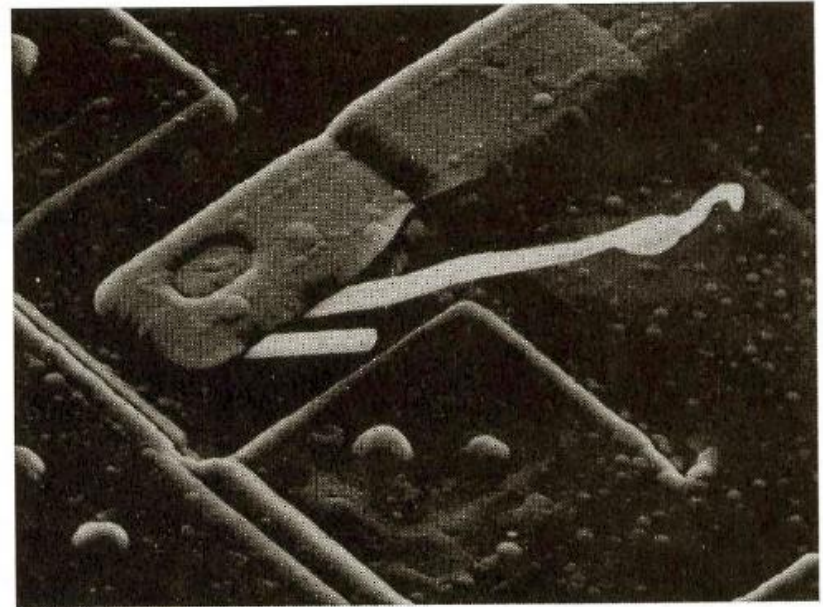
Transformata Fourier [1]

Filtrarea imaginilor in domeniul frecventelor – exemple:



Negativul imaginii:

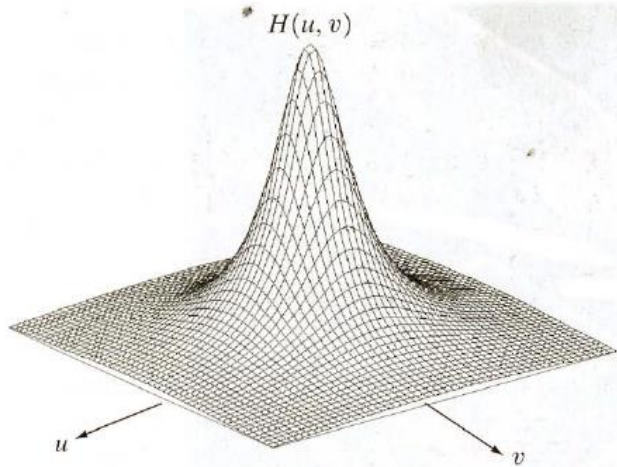
$$H(u, v) = \begin{cases} 0 & \text{if } (u, v) = (M/2, N/2) \\ 1 & \text{otherwise.} \end{cases}$$



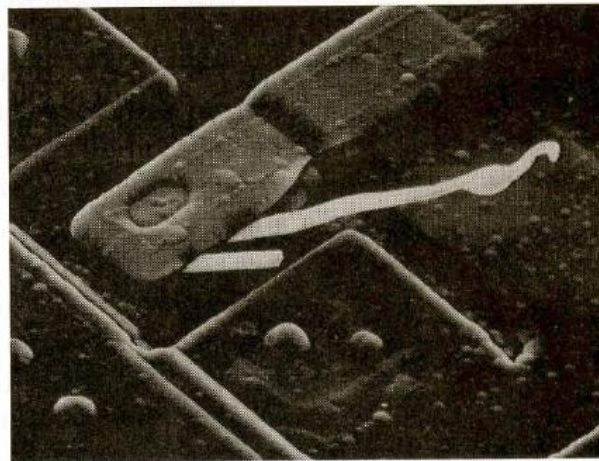
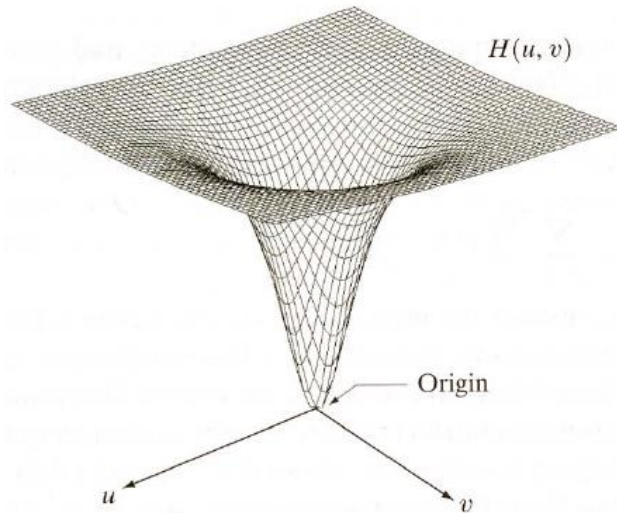


Transformata Fourier [1]

Filtrarea imaginilor in domeniul frecventelor – exemple:



FTJ

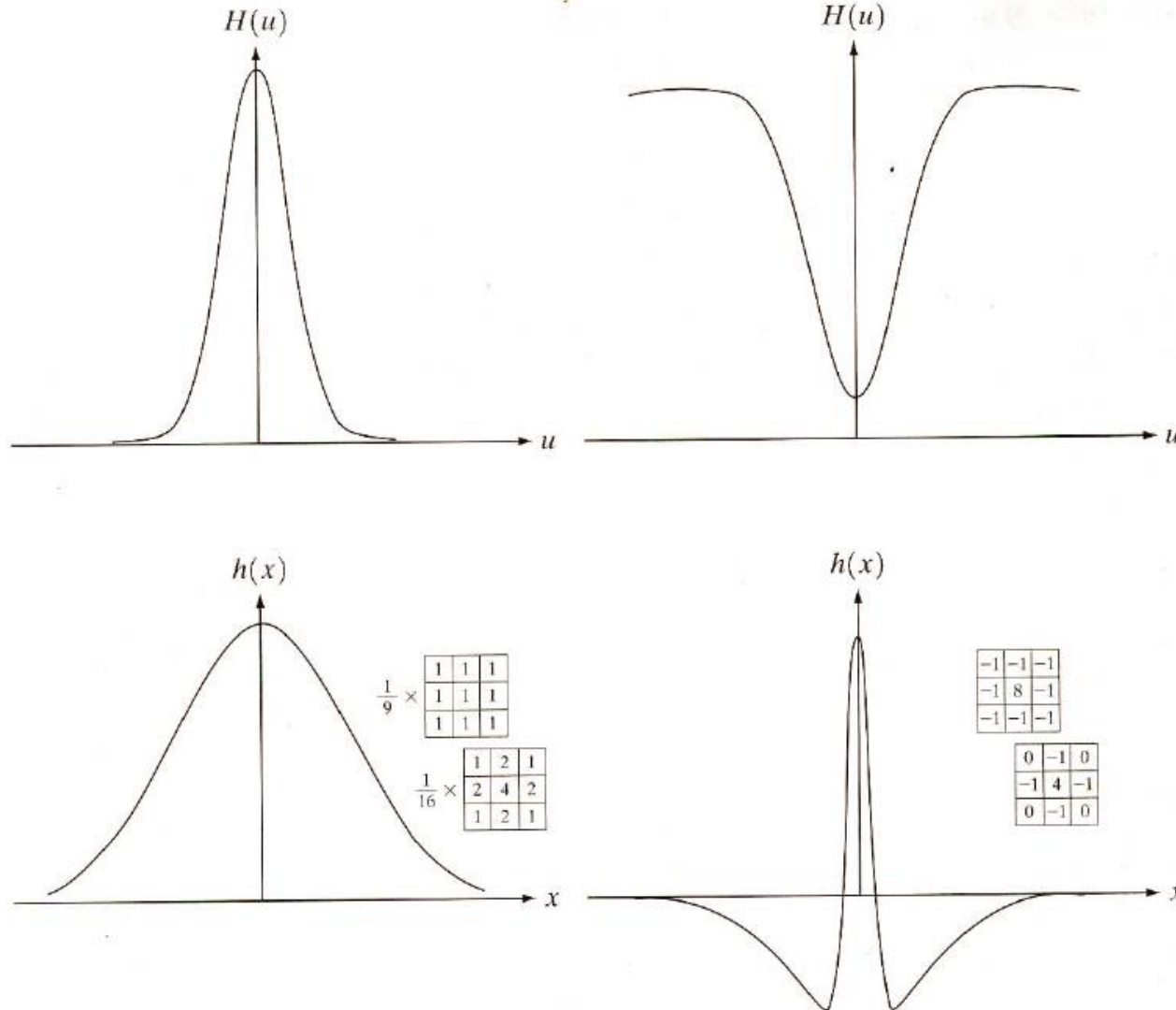


FTS



Transformata Fourier [1]

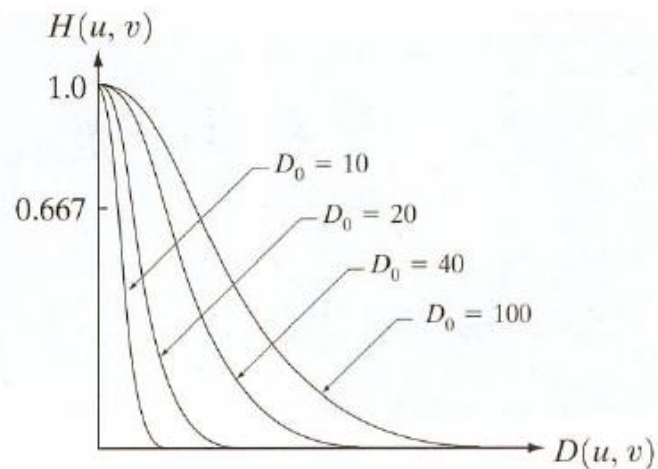
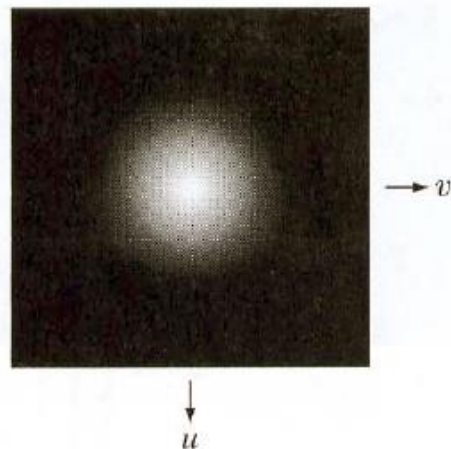
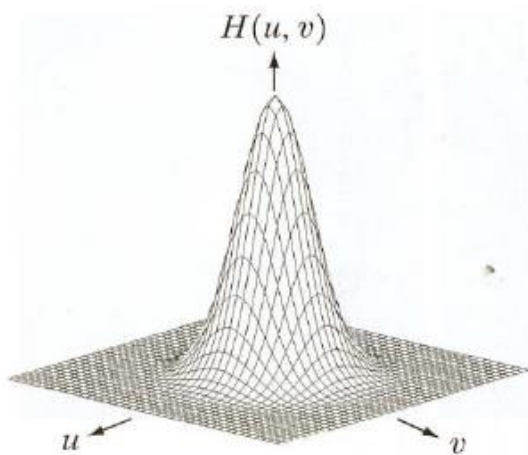
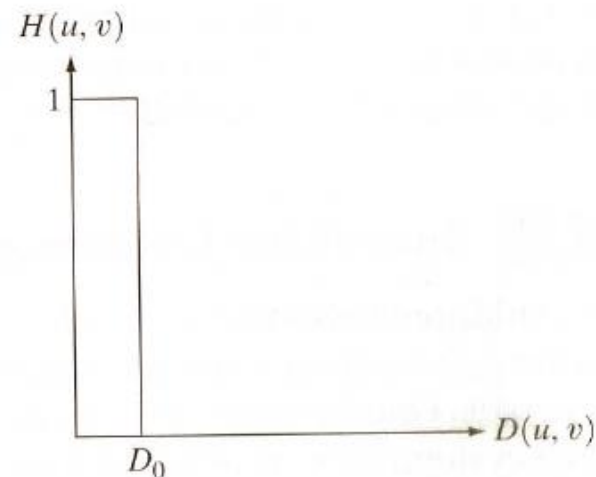
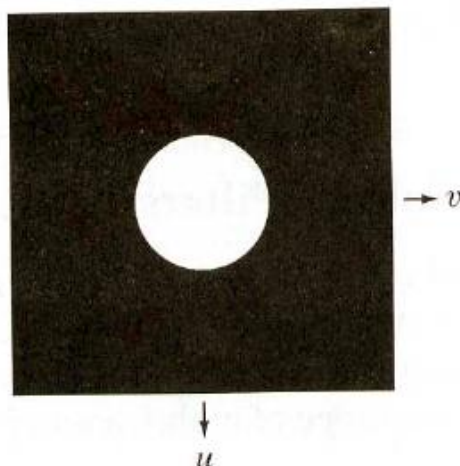
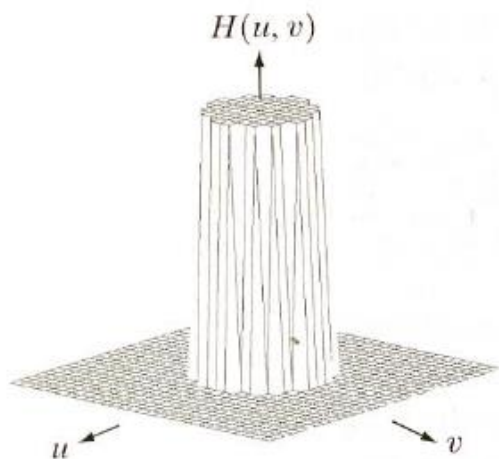
Filtrarea imaginilor in domeniul frecventelor – exemple:





Transformata Fourier [1]

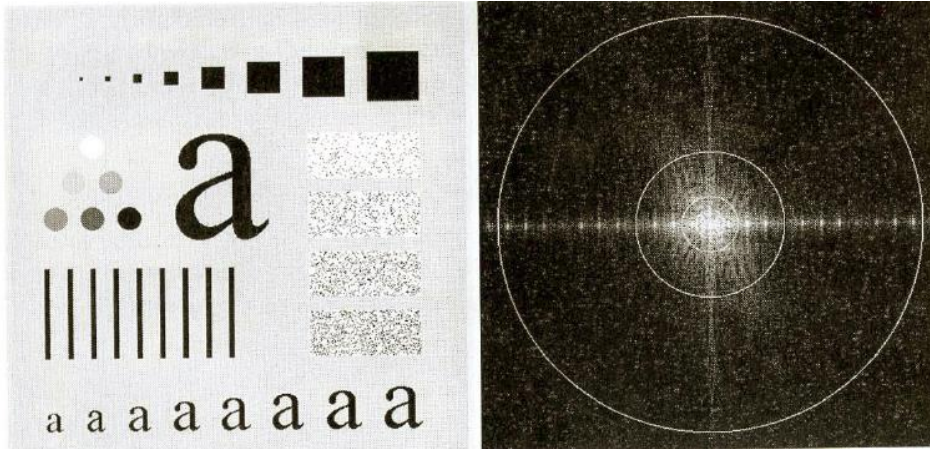
Filtrarea imaginilor in domeniul frecventelor – exemple: FTJ





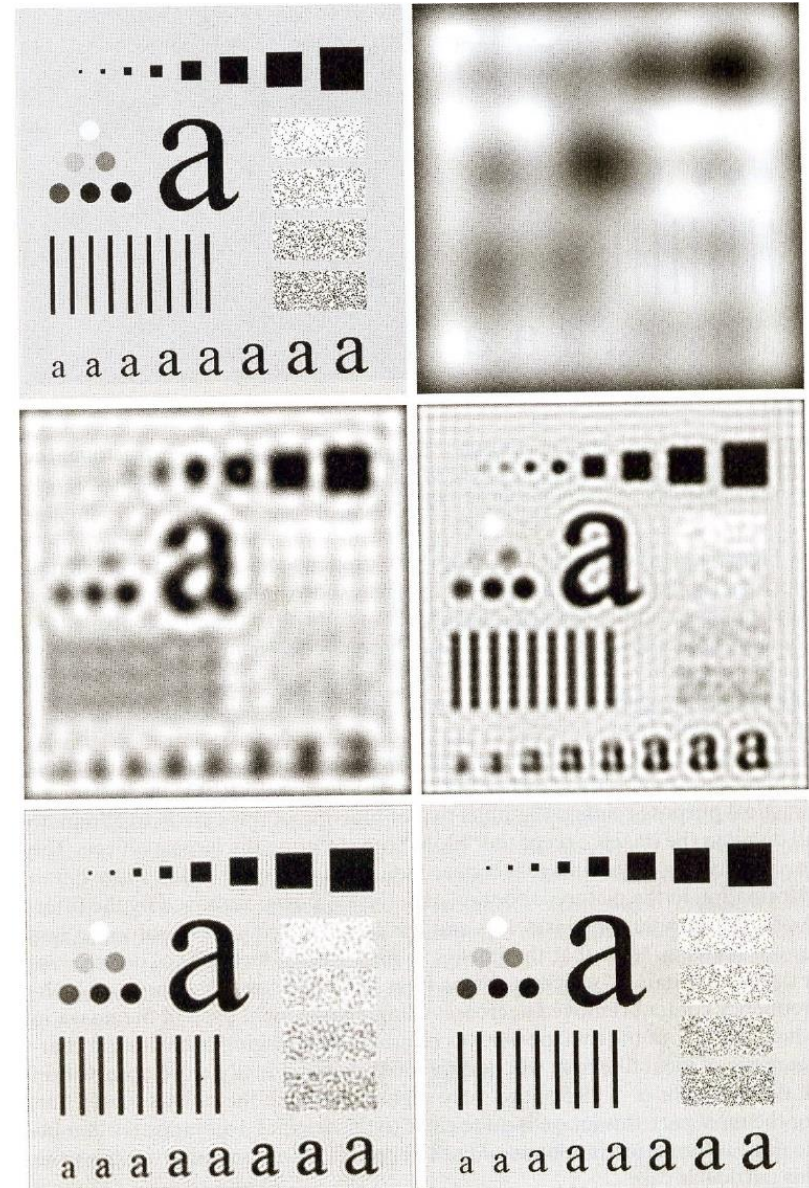
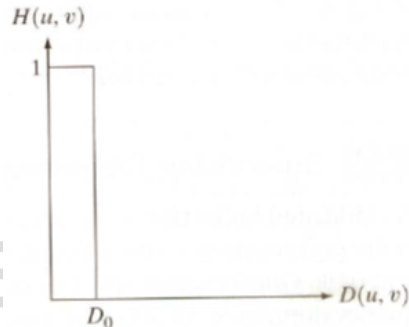
Transformata Fourier [1]

Filtrarea imaginilor in domeniul frecventelor – exemple: FTJ ideal



$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$$

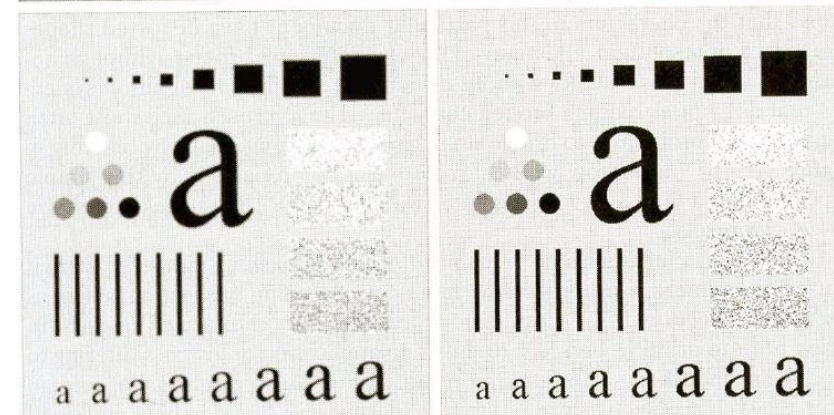
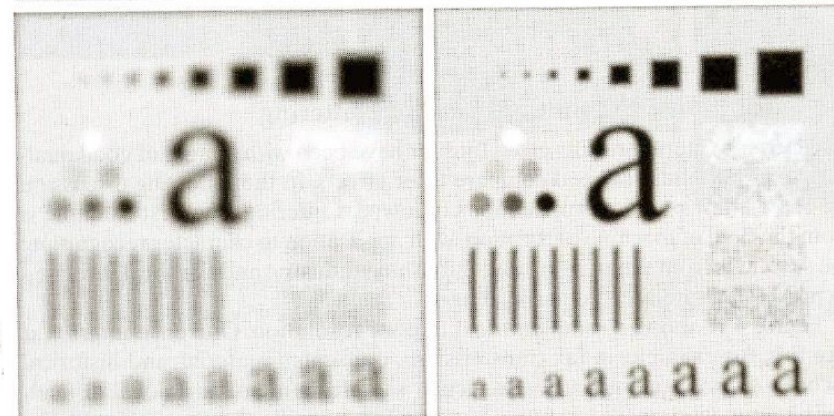
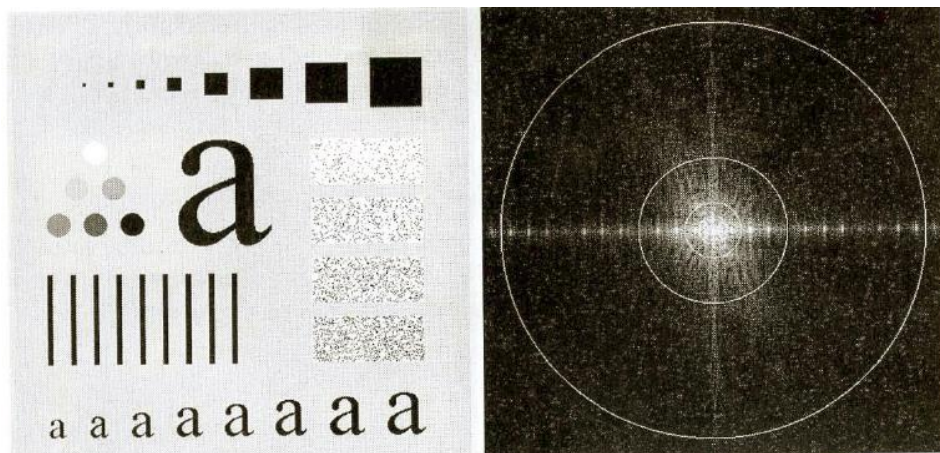
$$D(u, v) = [(u - M/2)^2 + (v - N/2)^2]^{1/2}$$





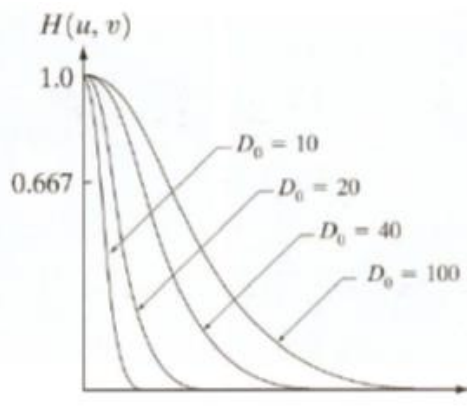
Transformata Fourier [1]

Filtrarea imaginilor in domeniul frecventelor – exemple: FTJ Gaussian



$$H(u, v) = e^{-D^2(u, v)/2D_0^2}$$

$$D(u, v) = [(u - M/2)^2 + (v - N/2)^2]^{1/2}$$





- [1] R.C.Gonzales, R.E.Woods, "Digital Image Processing, 2-nd Edition", *Prentice Hall*, 2002, pp 147-177